Project Design (Updated)

AI Medical Pre-Screening Assistant

EHR-Based Pre-Consultation Medical Documentation System

Rafiur Rahman Mashrafi (2221971042) M. G. Rabbi Hossen (2222516042) Israt Zaman Srity (2211084042)

Advisor: Dr. Mohammad Ashrafuzzaman Khan [AzK]

Dept. of ECE, North South University, Bangladesh

CSE499A — Spring 2026

1 Introduction

This chapter presents the **updated** design of the *AI Medical Pre-Screening Assistant*. The original CSE499A design was a five-stage, one-shot pipeline: patient speech → ASR → Medical NER → Structured EHR → Doctor Dashboard. Based on empirical findings from Phases 2 and 3 and a review of clinical NLP requirements, the design has been expanded to **15 functional modules** that add: an adaptive conversational follow-up loop, emergency detection, risk classification, explainable AI, and a continuous feedback and learning mechanism. The system remains a *pre-screening and decision-support tool*; all final clinical decisions belong to the attending physician.

2 Design Comparison: Original vs. Updated

Dimension	Original (5 stages)	Updated (15 modules)
Interaction model	One-shot recording; no follow-up	Conversational loop; adaptive follow-up until profile is complete
Emergency detection	Absent	Module 5: rule-based + AI classifier; immediate alert pathway
Information gaps	Absent	Module 6: structured completeness checklist per symptom category
Risk classification	Differential hints only	Module 10: Low / Medium / High / Critical with clinical rule engine
Explainability	Absent	Module 11 (XAI): plain-language driver explanations for every risk score
Clinical report	Basic structured EHR	Module 12: full 8-section pre-screening report; PDF + dashboard export
Feedback & learning	Absent	Module 15: doctor annotations feed model retraining pipeline
Bangla support	ASR & transcript only	End-to-end bilingual; follow-up questions generated in Bangla or English

3 Updated System Architecture — 15 Modules

3.1 Complete Pipeline Diagram

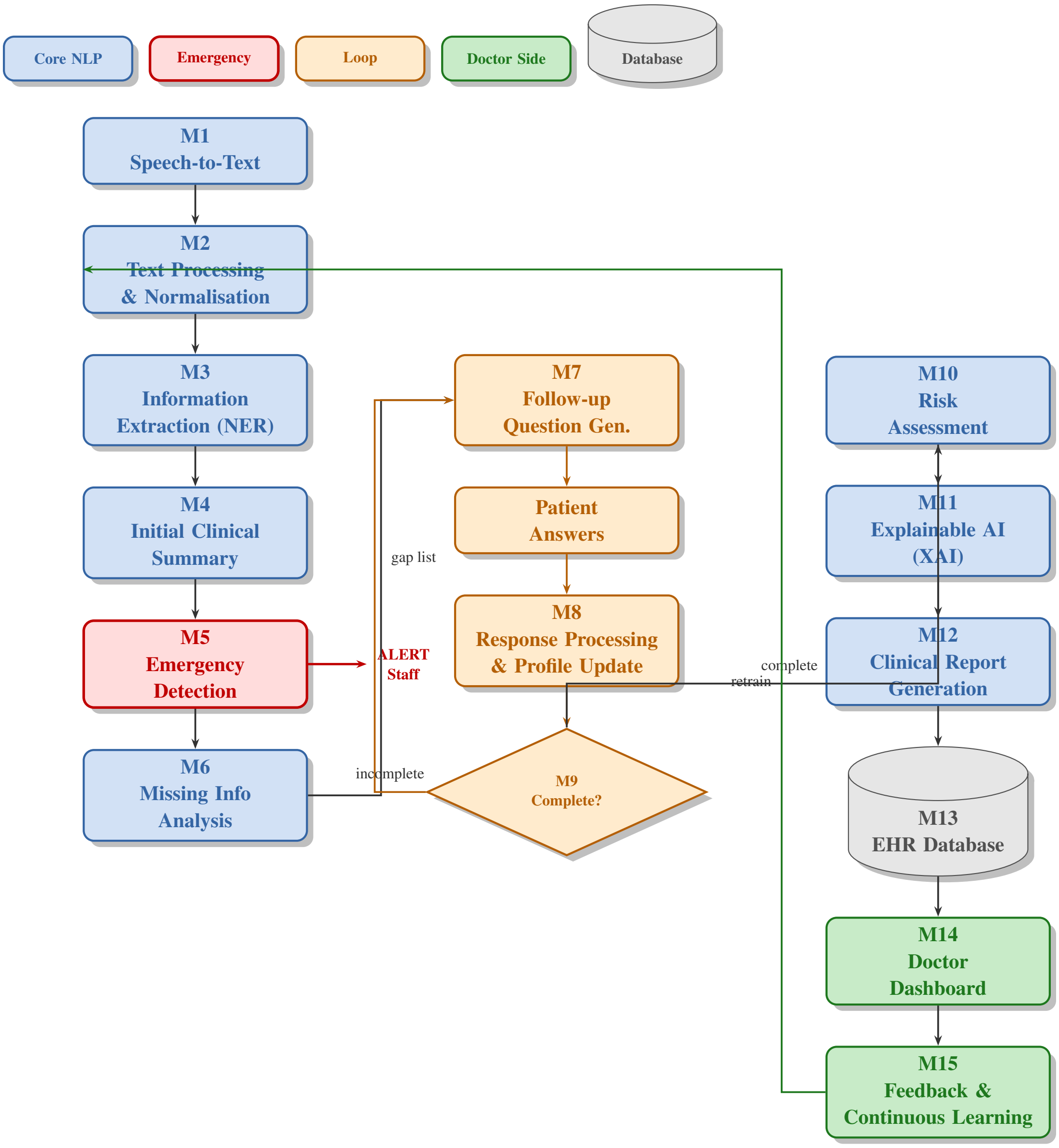

Figure 1: Complete 15-module pipeline. Orange = adaptive follow-up loop; red = emergency escalation path; green = doctor-facing modules.

3.2 Module Descriptions

M1 — Speech-to-Text.

Accepts Bangla, Banglish, and regional dialect speech. Audio is VAD-segmented into 10–15 s clips and transcribed by the best-performing ASR model identified in Phase 2 (*BengaliAI Regional*: 46.94% WER on dialect; target for CSE499B fine-tuning: $<$20% WER via LoRA on Qwen3-ASR-1.7B).

M2 — Text Processing & Normalisation.

Spelling correction, Banglish normalisation, filler removal, punctuation restoration, and sentence-boundary detection. Bilingual Bangla–English handling throughout.

M3 — Information Extraction (NER).

BanglaBERT token classifier labels: *symptom, affected site, duration, severity, frequency, age, comorbidity, medication*. Temporal expressions normalised to ISO 8601-like intervals.

M4 — Initial Clinical Summary.
Short human-readable summary of the chief complaint, e.g. *"Fever and dry cough for 3 days; mild fatigue; no dyspnoea or chest pain."* Used downstream and displayed on the dashboard.

M5 — Emergency Detection (new).
Rule-based critical-symptom checker + AI classifier for ambiguous presentations. Red-flag triggers: chest pain, stroke indicators, severe dyspnoea, loss of consciousness, acute trauma. Escalates immediately; bypasses the normal pipeline flow.

M6 — Missing Information Analysis (new).
Compares extracted entities against a per-symptom clinical checklist and outputs a structured gap report (e.g. ☑ Fever detected; ☒ Duration not specified; ☒ Temperature not provided).

M7 — Follow-up Question Generation (new).
Generates clinically prioritised questions in Bangla or English to fill identified gaps. Avoids repeating answered items. Example: *"Apnar jwor ki matro kal theke?"* / *"When did the fever start?"*

M8 — Response Processing & Profile Update (new).
Patient answers re-enter Modules 2–3. Extracted entities are merged into the profile with conflict resolution; filled gaps are marked.

M9 — Case Completion Check (new).
Scores profile completeness against a configurable threshold. If incomplete: loops back to M7. If complete (or max turns reached): continues to M10.

M10 — Risk Assessment (new).
Assigns one of four tiers — **Low / Medium / High / Critical** — using a clinical rule engine weighted by age, comorbidities, severity, and duration. Integrates M5 output.

M11 — Explainable AI (XAI) (new).
SHAP/LIME feature attribution converted to plain language: e.g. *"High risk: fever >5 days + dyspnoea + age >65."* Displayed alongside every risk classification.

M12 — Clinical Report Generation (new).
Produces an 8-section pre-screening report: Patient Profile, Chief Complaint, Symptoms, Emergency Flags, Q&A Log, Risk Level + XAI, Possible Condition Categories, Recommended Next Steps. Exported as PDF and to the dashboard.

M13 — EHR Database.
HIPAA/PDPA-compliant PostgreSQL store with encryption at rest and in transit. Stores transcripts, extracted data, follow-up exchanges, risk assessments, and final reports. Full audit logging.

M14 — Doctor Dashboard (extended from original).
Displays the report, risk tier, emergency flags, and XAI explanation. Supports annotation and override of any extracted field (all changes versioned). Real-time notifications for High/Critical cases.

M15 — Feedback & Continuous Learning (new).
Doctors rate reports, correct extractions, flag missed symptoms, and annotate risk disagreements. Feedback drives periodic model retraining for M3, M7, and M10, subject to regression testing before deployment.

4 System Flowchart — Patient Journey

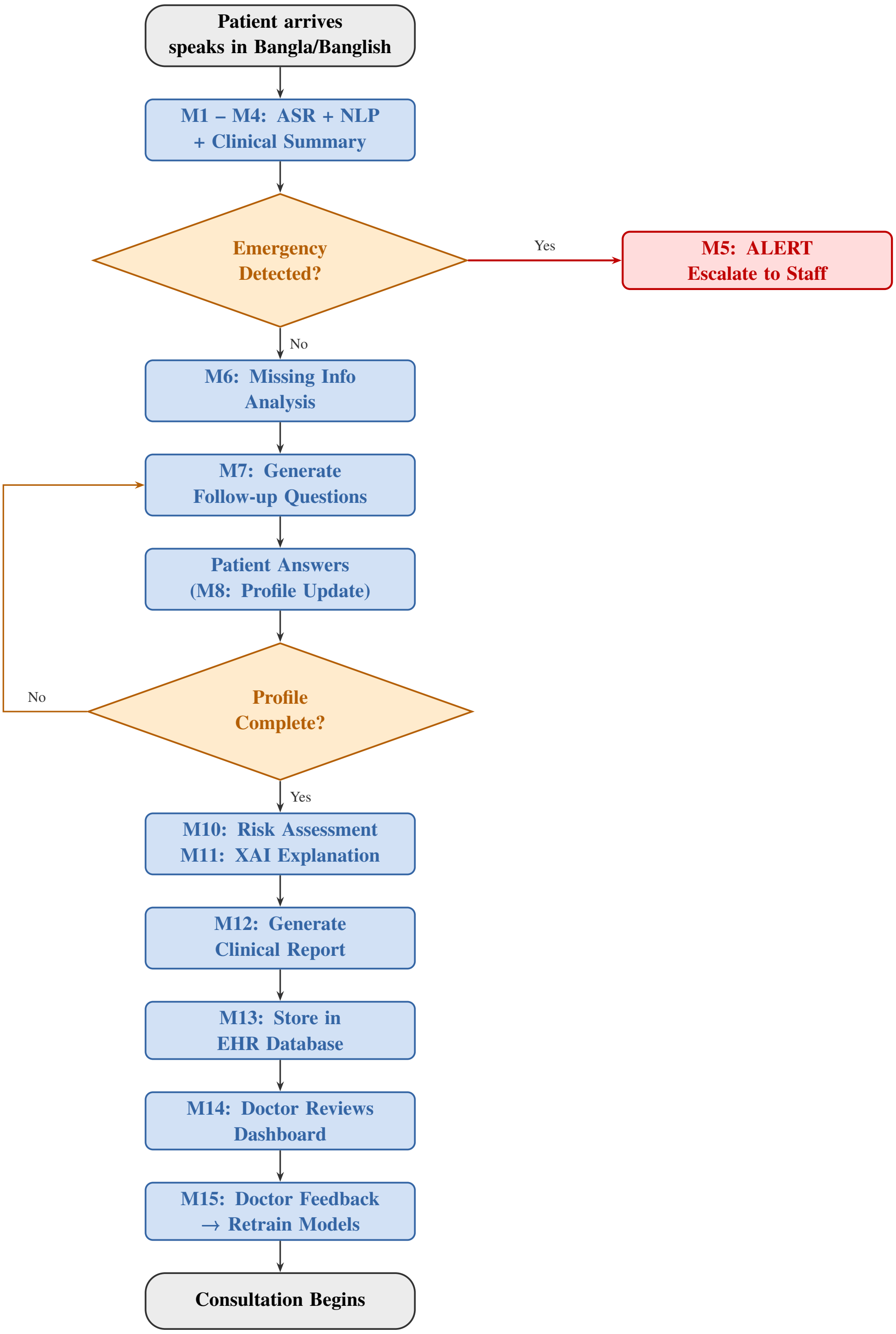

Figure 2: Patient journey flowchart through all 15 modules. The orange loop iterates until the clinical profile is sufficiently complete.

5 Software Components

Tool	Function	Rationale
Python 3.10	Implementation language	Standard ML/NLP ecosystem
PyTorch 2.x	Deep-learning framework	Native support across all benchmarked ASR models
HF Transformers	Model loading & inference	Unified API: Whisper, Wav2Vec2, MMS, Qwen-Audio, Phi-4
HF PEFT / Accelerate	LoRA fine-tuning (CSE499B)	Lowest friction with Transformers for parameter-efficient tuning
BanglaBERT	Medical NER (M3)	Pre-trained on large Bangla corpus; best Bangla NLP results
WhisperX	VAD segmentation (M1)	Required for audio >30 s; forced alignment
LIME / SHAP	XAI feature attribution (M11)	Standard, well-documented explainability libraries
FastAPI	Backend REST/WebSocket (M8–M13)	Async; low latency for real-time follow-up dialogue
PostgreSQL	EHR database (M13)	Mature; encryption-at-rest support; HIPAA-ready
React / Next.js	Doctor dashboard (M14)	Low-bandwidth responsive UI; role-based access
Flutter / PWA	Patient kiosk app (M1 interface)	Low-tech-literacy UX; single record button; Bangla prompts
MLflow	Feedback & retraining (M15)	Experiment tracking; model versioning
`jiwer`	WER / CER / BLEU scoring	Standard ASR evaluation metric library
Google Colab	GPU compute (T4 / A100)	Free tier sufficient for inference benchmarks

6 CSE499A Implementation Phases

Phase 1 — Audio Collection & Preprocessing
Demo: 12 Mar 2026. Multi-dialect corpus assembled (Dhakaiya, Barishali, Sylheti, Normal, Indian Bangla). Files logged in `dataset_log.csv`. Preprocessed to 16 kHz mono PCM; segmented by WhisperX VAD. Final corpus: $\approx$4.7 h, 10 sources, $\sim$1,100 clips.

Phase 2 — Baseline ASR Benchmark (12 models)
Demo: 2 Apr 2026. Whisper (S/M/L/Turbo), Wav2Vec2 variants, MMS, Vakyansh-Bn, BengaliAI Whisper/Regional, SeamlessM4T. Best: **BengaliAI Regional** (46.94% WER, 24.29% CER, 0.273 BLEU). Finding: dialect speech defeats even the best open Bangla models.

Phase 3 — Large Multimodal LLM Benchmark
Demo: 9 Apr 2026. Qwen2-Audio, Qwen3-ASR-1.7B, Voxtral-Mini, Phi-4 Multimodal, Qwen2.5-Omni (1.7B–7B). None matched BengaliAI Regional; closest trailed by $>$50 pp WER. Conclusion: *specialisation beats scale for Bangla dialect ASR*.

Phase 4 — Fine-Tuning Research (in progress)
Demo: 16 Apr 2026. LoRA fine-tuning methodology for Qwen3-ASR-1.7B on the dialect corpus. Adapter on speech encoder only; decoder frozen. No training run yet — this is a methodology study for execution in CSE499B.

Phase 5 — Chatbot Output Study (selected)
Execution: next demo. ChatGPT, DeepSeek, Qwen, Perplexity prompted with simulated patient utterances.

Outputs compared on schema consistency, field completeness, code-mixed handling, and source faithfulness. Result will inform Module 12 report schema.

7 Conclusion

The updated design addresses the three main limitations of the original five-stage pipeline: (i) the adaptive follow-up loop (M6–M9) ensures clinical information completeness; (ii) Module 5 provides an explicit emergency escalation pathway absent from the original design; and (iii) Modules 10–11 and 15 add risk stratification, explainability, and continuous improvement. The empirical finding from Phases 2 and 3 — that specialisation beats scale for Bangla dialect ASR — directly motivates the CSE499B priority: LoRA fine-tuning of Qwen3-ASR-1.7B. Once ASR reliability is demonstrated, the application layer (Modules 3–15) will be built iteratively.

References

[1] A. Radford *et al.*, "Robust Speech Recognition via Large-Scale Weak Supervision," *ICML 2023*. `https://arxiv.org/abs/2212.04356`

[2] A. Baevski *et al.*, "wav2vec 2.0," *NeurIPS 2020*.

[3] Y. Chu *et al.*, "Qwen2-Audio Technical Report," *arXiv:2407.10759*, 2024.

[4] T. Wolf *et al.*, "Transformers," *EMNLP 2020*. `https://github.com/huggingface/transformers`

[5] E. J. Hu *et al.*, "LoRA," *ICLR 2022*. `https://arxiv.org/abs/2106.09685`

[6] M. Bain *et al.*, "WhisperX," *INTERSPEECH 2023*. `https://arxiv.org/abs/2303.00747`

[7] A. Bhattacharjee *et al.*, "BanglaBERT," *NAACL 2022 Findings*. `https://github.com/csebuetnlp/banglabert`

[8] S. Lundberg & S. Lee, "SHAP," *NeurIPS 2017*. `https://arxiv.org/abs/1705.07874`

[9] M. Ribeiro *et al.*, "LIME," *KDD 2016*. `https://arxiv.org/abs/1602.04938`

[10] F. R. Rakib *et al.*, "OOD-Speech," *INTERSPEECH 2023*. `https://arxiv.org/abs/2305.09688`